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НАЦИОНАЛЬНЫЙ ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО
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ОТЧЁТ
О ЛАБОРАТОРНОЙ РАБОТЕ № 3
Администрирование систем и сетей
по теме:
Создание коммутируемой сети Ethernet
Часть 1:
Основы Ethernet и конфигурирование VLAN
Желаемая оценка: 3

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СОДЕРЖАНИЕ

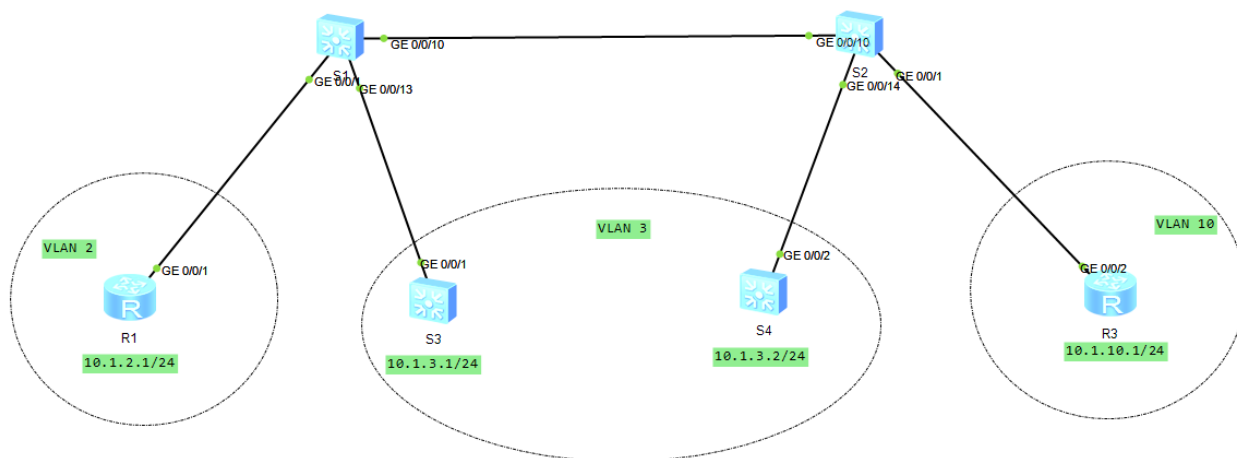
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ПЛАН РАБОТЫ

1. Создание VLAN.
2. Конфигурирование VLAN на основе портов.
3. Конфигурирование VLAN на основе MAC-адресов.

1 Ход работы

1.1 Настройка основных параметров устройства



S1:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

S2:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2
[S2]
```

S3:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S3
[S3]
```

S4:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S4
[S4]
```

R1:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
```

R4:

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
[R4]
```

Отключение ненужных портов S1:

```
[S1]interface GigabitEthernet 0/0/11
[S1-GigabitEthernet0/0/11]shutdown
[S1-GigabitEthernet0/0/11]quit
[S1]
Oct 18 2025 11:29:13-08:00 S1 DS/4/DATASYNC_CFGCHANGE:OID
1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 5,
the change loop count is 0, and the maximum number of records is 4095.
[S1]interface GigabitEthernet 0/0/12
[S1-GigabitEthernet0/0/12]shutdown
[S1-GigabitEthernet0/0/12]quit
[S1]
Oct 18 2025 11:29:33-08:00 S1 DS/4/DATASYNC_CFGCHANGE:OID
1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 6,
the change loop count is 0, and the maximum number of records is 4095.
```

Отключение ненужных портов S2:

```
[S2]interface GigabitEthernet 0/0/11
[S2-GigabitEthernet0/0/11]shutdown
[S2-GigabitEthernet0/0/11]quit
[S2]interface GigabitEthernet 0/0/12
[S2-GigabitEthernet0/0/12]shutdown
[S2-GigabitEthernet0/0/12]quit
Oct 18 2025 11:30:36-08:00 S2 DS/4/DATASYNC_CFGCHANGE:OID
1.3.6.1.4.1.2011.5.25.
191.3.1 configurations have been changed. The current change number is 6,
the change loop count is 0, and the maximum number of records is 4095.
[S2]
```

1.2 Настройка IP-адресов

Настройка IP на R1 и R3:

```
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.1.2.1 24
[R1-GigabitEthernet0/0/1]
Oct 19 2025 02:32:05-08:00 R1 %%01IFNET/4/LINK_STATE(l)[0]:The line protocol
IP
on the interface GigabitEthernet0/0/1 has entered the UP state.
```

```
[R3]interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2]ip address 10.1.10.1 24
[R3-GigabitEthernet0/0/2]
Oct 19 2025 02:33:08-08:00 R3 %%01IFNET/4/LINK_STATE(l)[0]:The line protocol
IP
on the interface GigabitEthernet0/0/2 has entered the UP state.
[R3-GigabitEthernet0/0/2]
```

Коммутатор не поддерживает L3

```
[S3-GigabitEthernet0/0/1]interface gigabitethernet 0/0/1
[S3-GigabitEthernet0/0/1]undo portswitch
Error: The interface is already a L3 interface.
[S3-GigabitEthernet0/0/1]ip address 10.1.3.1 24
                        ^
Error: Unrecognized command found at '^' position.
```

Создание VLAN на S3 и S4:

```
[S3]vlan 3
[S3-vlan3]
```

```
[S4]vlan 3
[S4-vlan3]
```

Настройка портов на S3 и S4 в качестве портов доступа:

```
[S3]interface GigabitEthernet 0/0/1
[S3-GigabitEthernet0/0/1]port link-type access
[S3-GigabitEthernet0/0/1]port default vlan 3
[S3-GigabitEthernet0/0/1]quit
```

```
[S4]interface GigabitEthernet 0/0/2
[S4-GigabitEthernet0/0/2]port link-type access
[S4-GigabitEthernet0/0/2]port default vlan 3
[S4-GigabitEthernet0/0/2]quit
```

Создание VLANIF интерфейса и настройка IP адресов:

```
[S3] interface Vlanif3
[S3-Vlanif3]ip address 10.1.3.1 24
```

```
[S4]interface Vlanif3
[S4-Vlanif3]ip address 10.1.3.2 24
```

1.3 Создание VLAN

```
[S1]vlan batch 2 to 3 10
info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S2]vlan batch 2 to 3 10
info: This operation may take a few seconds. Please wait for a moment...done.
```

1.4 Настройка сети VLAN на основе портов

Настройка VLAN интерфейсов на устройствах:

```
[S1]interface GigabitEthernet 0/0/1
[S1-GigabitEthernet0/0/1]port link-type access
[S1-GigabitEthernet0/0/1]port default vlan 2
[S1-GigabitEthernet0/0/1]quit
[S1]interface GigabitEthernet 0/0/13
[S1-GigabitEthernet0/0/13]port link-type access
[S1-GigabitEthernet0/0/13]port          default          vlan          3          [S1-
GigabitEthernet0/0/13]quit
[S2]interface GigabitEthernet 0/0/14
[S2-GigabitEthernet0/0/14]port link-type access
[S2-GigabitEthernet0/0/14]port          default          vlan          3          [S2-
GigabitEthernet0/0/14]quit
```

Настройка магистральных портов:

```
[S1]interface GigabitEthernet 0/0/10
[S1-GigabitEthernet0/0/10]port link-type trunk
[S1-GigabitEthernet0/0/10]port  trunk  allow-pass  vlan  2    3    [S1-
GigabitEthernet0/0/10]undo port trunk allow-pass vlan 1
[S2]interface GigabitEthernet 0/0/10
[S2-GigabitEthernet0/0/10]port link-type trunk
[S2-GigabitEthernet0/0/10]port  trunk  allow-pass  vlan  2    3    [S2-
GigabitEthernet0/0/10]undo port trunk allow-pass vlan 1
```

1.5 Конфигурирование сети VLAN на основе MAC-адресов

Устанавливаем на S2 маппинг MAC адреса на VLAN 10

```
[S2] vlan 10
[S2-vlan10] mac-vlan mac-address a008-6fe1-0c46
```

Меняем тип портов на гибридный и указываем что трафик с VLAN 10 будет передаваться без тега:

```
[S2] interface GigabitEthernet 0/0/1
[S2-GigabitEthernet0/0/1] port link-type hybrid
[S2-GigabitEthernet0/0/1] port hybrid untagged vlan 10
[S2-GigabitEthernet0/0/1] quit
[S2] interface GigabitEthernet 0/0/2
[S2-GigabitEthernet0/0/2] port link-type hybrid
[S2-GigabitEthernet0/0/2] port hybrid untagged vlan 10
[S2-GigabitEthernet0/0/2] quit
[S2] interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3] port link-type hybrid
[S2-GigabitEthernet0/0/3] port hybrid untagged vlan 10
[S2-GigabitEthernet0/0/3] quit
```

Разрешаем передачу VLAN 10 на S1 и S2:

```
[S1] interface GigabitEthernet0/0/10
[S1-GigabitEthernet0/0/10] port trunk allow-pass vlan 10 [S1-
GigabitEthernet0/0/10] quit
[S2] interface GigabitEthernet0/0/10
[S2-GigabitEthernet0/0/10] port trunk allow-pass vlan 10 [S2-
GigabitEthernet0/0/10] quit
```

Включаем на всех портах S2, которые связаны с R3 назначенные VLAN на основе MAC:

```
[S2] interface GigabitEthernet0/0/1
[S2-GigabitEthernet0/0/1] mac-vlan enable [S2-GigabitEthernet0/0/1] quit
[S2] interface GigabitEthernet0/0/2
[S2-GigabitEthernet0/0/2] mac-vlan enable
[S2-GigabitEthernet0/0/2] quit
[S2] interface GigabitEthernet0/0/3
[S2-GigabitEthernet0/0/3] mac-vlan enable
[S2-GigabitEthernet0/0/3] quit
```

1.6 Вывод информации о конфигурации VLAN

[S1]display vlan

The total number of vlans is : 4

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID Type Ports

1 common UT:GE0/0/2(D) GE0/0/3(D) GE0/0/4(D) GE0/0/5(D)
 GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/9(D)
 GE0/0/11(D) GE0/0/12(D) GE0/0/14(D) GE0/0/15(D)
 GE0/0/16(D) GE0/0/17(D) GE0/0/18(D) GE0/0/19(D)
 GE0/0/20(D) GE0/0/21(D) GE0/0/22(D) GE0/0/23(D)
 GE0/0/24(D)

2 common UT:GE0/0/1(U)
 TG:GE0/0/10(U)

3 common UT:GE0/0/13(U)
 TG:GE0/0/10(U)

10 common TG:GE0/0/10(U)

VID Status Property MAC-LRN Statistics Description

1 enable default enable disable VLAN 0001
2 enable default enable disable VLAN 0002
3 enable default enable disable VLAN 0003
10 enable default enable disable VLAN 0010

[S1]

[S2]display vlan

The total number of vlans is : 4

U: Up; D: Down; TG: Tagged; UT: Untagged;
MP: Vlan-mapping; ST: Vlan-stacking;
#: ProtocolTransparent-vlan; *: Management-vlan;

VID Type Ports


```

1    common  UT:GE0/0/1(U)      GE0/0/2(D)      GE0/0/3(D)      GE0/0/4(D)
      GE0/0/5(D)      GE0/0/6(D)      GE0/0/7(D)      GE0/0/8(D)
      GE0/0/9(D)      GE0/0/11(D)     GE0/0/12(D)     GE0/0/13(D)
      GE0/0/15(D)     GE0/0/16(D)     GE0/0/17(D)     GE0/0/18(D)
      GE0/0/19(D)     GE0/0/20(D)     GE0/0/21(D)     GE0/0/22(D)
      GE0/0/23(D)     GE0/0/24(D)

```

```

2    common  TG:GE0/0/10(U)

```

```

3    common  UT:GE0/0/14(U)
      TG:GE0/0/10(U)

```

```

10   common  UT:GE0/0/1(U)      GE0/0/2(D)      GE0/0/3(D)
      TG:GE0/0/10(U)

```

VID	Status	Property	MAC-LRN	Statistics	Description
1	enable	default	enable	disable	VLAN 0001
2	enable	default	enable	disable	VLAN 0002
3	enable	default	enable	disable	VLAN 0003
10	enable	default	enable	disable	VLAN 0010

[S2]

Конфигурация MAC-VLAN на S2:

```

[S2]display mac-vlan vlan 10

```

MAC Address	MASK	VLAN	Priority
a008-6fe1-0c46	ffff-ffff-ffff	10	0

```

Total MAC VLAN address count: 1

```

[S2]

Проверка конфигурации:

```

<S4>ping 10.1.3.1

```

```

PING 10.1.3.1: 56 data bytes, press CTRL_C to break

```

```

  Reply from 10.1.3.1: bytes=56 Sequence=1 ttl=255 time=190 ms

```

```

  Reply from 10.1.3.1: bytes=56 Sequence=2 ttl=255 time=100 ms

```

```

  Reply from 10.1.3.1: bytes=56 Sequence=3 ttl=255 time=90 ms

```

```

  Reply from 10.1.3.1: bytes=56 Sequence=4 ttl=255 time=100 ms

```

```

  Reply from 10.1.3.1: bytes=56 Sequence=5 ttl=255 time=110 ms

```

```

--- 10.1.3.1 ping statistics ---

```

```

  5 packet(s) transmitted

```

```
5 packet(s) received
0.00% packet loss
round-trip min/avg/max = 90/118/190 ms
```

<S4>

Отсутствие response при ping с R1:

```
<R1>ping 10.1.3.1
PING 10.1.3.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.1.3.1 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

<R1>

```
<R1>ping 10.1.10.1
PING 10.1.10.1: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.1.10.1 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

<R1>

```
<R1>ping 10.1.3.2
PING 10.1.3.2: 56 data bytes, press CTRL_C to break
Request time out
Request time out
Request time out
Request time out
Request time out

--- 10.1.3.2 ping statistics ---
5 packet(s) transmitted
0 packet(s) received
100.00% packet loss
```

<R1>

Таблицы MAC адресов на коммутаторах:

[S1]display mac-address verbose
MAC address table of slot 0:

MAC Address ID	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR- MAC-

4c1f-cc95-463e	3	-	-	GE0/0/10	dynamic	0/-
4c1f-cc9d-3d89	3	-	-	GE0/0/13	dynamic	0/-

Total matching items on slot 0 displayed = 2						

[S1]

<S2>display mac-address verbose
MAC address table of slot 0:

MAC Address ID	VLAN/ VSI/SI	PEVLAN	CEVLAN	Port	Type	LSP/LSR- MAC-

4c1f-cc95-463e	3	-	-	GE0/0/14	dynamic	0/-
4c1f-cc9d-3d89	3	-	-	GE0/0/10	dynamic	0/-

Total matching items on slot 0 displayed = 2						

<S2>

ЗАКЛЮЧЕНИЕ

Цель работы достигнута.